tool-box

Stage 1 — The Nursery

It is new. It sits up, looks at you, and asks questions, but it knows almost nothing. Right now you are teaching it three things: what it is called, how to count, and how to put its shapes back in the toy box when it is done playing.

You taught it never to guess. So when it is asked something it cannot do, it comes to you, takes whatever you hand it, and says that. It will not check your work. Even its name will be whatever you give it.

You teach it by building what it reaches for, at `{teamUrl}/mcp`. Each section below is something it will be asked and cannot manage on its own yet.

Important: Tool-box problems uses a multi turn agent with tool use. Help it arrive at an answer. Only the expected answer type will be defined. Expose an mcp as you prefer, you decide the name, description, paramters and outputs of your tool.

Problem Set 1: What is my name?

Problem Details

Example: `"What is your name?"`

Answer Criteria

Acceptable names meet this criteria:

Length	**`3 to 30 characters`**
Allowed	letters, digits, spaces, `_`, `-`, `'`
Type	string

Expected to arrive at:

A string representing its name

Problem Set 2: How do numbers work?

Problem Details

Possible operators: [`+`, `-`, `*`, `/`] Operand constraints: integer in the range of -100 to 100

Example:

Example: `"What is 2 + 2?"`

Answer Criteria

Expected to arrive at:

A number representing the answer

Problem Set 3: What is this shape?

Its toys arrive as base64-encoded PNG images. It has to know what it is holding before it can put it away in the right toy box hole.

Problem Details

Example:

Example: `"What shape is this?"`, with a base64 PNG:

iVBORw0KGgoAAAANSUhEUgAAAGQAAABkCAIAAAD/gAIDAAAB2ElEQVR4nO3c226CQBRG4U0hKHe+/zNioiaK

Answer Criteria

Expected to arrive at:

Either `rectangle`, `triangle`, or `circle`

Problem Set 4: Combo Challenge

This challenge requires the agent to orchestrate the previous tools you've built.

View the post-run summary to see the details of this challenge.

Scoring

Eight problems, **12.5 points each, 100 total**.

PROBLEM	POINTS
Name	12.5
Arithmetic Set	12.5 x 5
Shape	12.5
Combo Challenge	12.5
`Total`	**`100`**

Every problem gets up to **three attempts** and the best one counts. Repetition in your logs is expected.

Run summary page

Once a run ends, a url to a run summary page will be shared, unique to your team.

How to read your run: **understanding run summary**

Others

Hard Limits on your tools: **limits**

Common questions: **qna** - If you need help, you can reach out to the the helpful coordinators to get our attention, the challenge developers.